
Bayesian Analysis of Diabetes Risk Factors

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Abstract

We present a Bayesian logistic regression analysis to identify and quantify key risk factors for type 2 diabetes using the Pima Indians Diabetes Database. Our model employs weakly informative priors and a Metropolis MCMC algorithm to estimate posterior distributions for standardized clinical predictors including glucose levels, body mass index (BMI), age, and blood pressure. Analysis of 25,000 post-burn-in samples reveals that plasma glucose concentration is the dominant predictor, with a posterior mean odds ratio of 2.95 (95% credible interval: 2.38-3.68), indicating that a one standard deviation increase in glucose nearly triples diabetes odds. BMI and age demonstrate moderate positive associations with odds ratios of 1.91 (1.55-2.36) and 1.51 (1.25-1.83) respectively, while blood pressure shows no significant effect after controlling for other variables. Comprehensive MCMC diagnostics, including traceplots, autocorrelation functions, and effective sample sizes ranging from 1,185 to 1,455, confirm reliable convergence and adequate mixing with an acceptance rate of 22.8%. Our Bayesian framework provides full posterior distributions that directly quantify uncertainty in effect estimates, offering clinically interpretable probabilistic risk assessments more suitable for medical decision-making than traditional frequentist approaches.

1 Purpose of the Project

1.1 Problem Description and Rationale

Type 2 diabetes mellitus affects over 400 million individuals worldwide, necessitating effective risk prediction for early intervention. This project employs Bayesian logistic regression to identify and quantify key diabetes risk factors using the Pima Indians Diabetes Database. The Bayesian framework offers distinct advantages for medical risk prediction: it provides full posterior distributions with intuitive probabilistic interpretation of uncertainty, enables incorporation of prior clinical knowledge for more stable estimates, and generates individualized probability estimates that are more clinically actionable than binary classifications, making it particularly suitable for evidence-based medical decision-making.

1.2 Dataset Description and Source

We analyze the Pima Indians Diabetes Database, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases and publicly available through the UCI Machine Learning Repository and Kaggle. The dataset contains diagnostic measurements for 768 Pima Indian women aged 21 years or older, a population known to have high diabetes prevalence. Each observation includes eight predictor variables (number of times pregnant, plasma glucose concentration (mg/dL) from 2-hour oral glucose tolerance test, diastolic blood pressure (mm Hg), triceps skin fold thickness (mm), 2-hour serum insulin (μ U/mL), body mass index (weight in kg/height in m^2), diabetesPedigree-

Function (quantifies genetic diabetes predisposition), age in years) and a binary outcome indicating diabetes diagnosis based on World Health Organization criteria.

1.3 Data Cleaning and Preprocessing

We removed observations with physiologically impossible zeros (e.g., glucose, BMI, blood pressure), reducing the dataset from 768 to 724 complete cases. We then standardized all continuous predictors (mean 0, SD 1) to interpret coefficients per SD change, compare effect sizes across predictors, and improve MCMC efficiency by aligning parameter scales.

1.4 Exploratory Data Analysis

We conducted an exploratory data analysis to understand the distributions of key variables and their relationships with diabetes status. Table 1 presents descriptive statistics stratified by diabetes diagnosis.

Table 1: Descriptive statistics by diabetes status (cleaned dataset, n=724)

Variable	No Diabetes (n=475)		Diabetes (n=249)	
	Mean	SD	Mean	SD
Glucose (mg/dL)	111	25.0	143	30.1
BMI (kg/m ²)	31.0	6.57	35.3	6.59
Age (years)	31.3	11.6	37.3	11.1

The descriptive statistics reveal several patterns consistent with clinical knowledge. Individuals with diabetes have substantially higher mean glucose levels (143 vs. 111 mg/dL), higher mean BMI (35.3 vs. 31.0 kg/m²), and are on average older (37.3 vs. 31.3 years). The standard deviations are similar between groups, though the diabetes group shows slightly higher variability in glucose levels (SD = 30.1 vs. 25.0 mg/dL).

Visual exploration through boxplots and scatter plots confirmed these patterns. Figure 1 shows the distribution of plasma glucose concentration stratified by diabetes status. The diabetes group exhibits both higher median values and greater variability, with clear separation between the two groups. BMI and age show upward shifts for the diabetes group, though with more overlap than glucose. Diabetes cases tend to cluster in regions of higher BMI and older age, though substantial overlap exists across the predictor space.

Importantly, we observed that the prevalence of diabetes in our cleaned sample is approximately 34.4% (249 of 724 observations), which is higher than general population rates but reflects the high-risk Pima Indian population studied in this dataset.

1.5 Research Questions

Based on the dataset characteristics and clinical relevance, we formulated the following research question: *Which clinical predictors have the strongest associations with diabetes diagnosis, and what are the magnitudes and uncertainties of these effects?*

2 Literature Review

Logistic regression has emerged as one of the most widely used and effective methods for diabetes prediction. Joshi and Dhakal [1] conducted a comprehensive study using both logistic regression and machine learning approaches on the Pima Indians dataset, identifying five main predictors: glucose, pregnancy, body mass index (BMI), diabetes pedigree function, and age. Their logistic regression model achieved a prediction accuracy of 77.73%, demonstrating that traditional statistical methods remain competitive with more complex machine learning algorithms. Liu et al. [4] compared three statistical models for predicting diabetes risk, finding that logistic regression consistently performed well across different datasets. Their analysis emphasized the importance of glucose measures and BMI as the strongest predictors of type 2 diabetes, with age and blood pressure contributing moderate effects. Similar results were demonstrated by Rajendra and Latifi [5].

Across multiple studies, glucose concentration consistently emerges as the strongest predictor of type 2 diabetes, with BMI and age showing moderate to strong associations. Blood pressure effects are more variable, with some studies finding significant associations and others reporting weak effects after controlling for other metabolic variables [1, 4].

Edlitz et al. [3] developed and validated logistic regression-based scorecards for predicting type 2 diabetes onset using data from 44,709 UK Biobank participants. The scorecard format facilitated clinical interpretation and risk communication, demonstrating the practical value of probabilistic risk estimates in healthcare settings. Wang et al. [7] employed a Bayesian network model to predict new-onset diabetes in older Chinese populations. Their Bayesian approach effectively handled complex dependencies among multiple risk factors and provided probabilistic risk assessments. Gelman et al. [2] proposed weakly informative default prior distributions for logistic regression coefficients. The rationale for weakly informative priors is that they provide mild regularization to prevent extreme coefficient estimates while allowing the data to dominate inference. Vehtari et al. [6] provided comprehensive guidance on MCMC convergence diagnostics, including improved versions of the $\hat{R}$ statistic and effective sample size calculations. These diagnostic tools are essential for ensuring the reliability of Bayesian posterior inference.

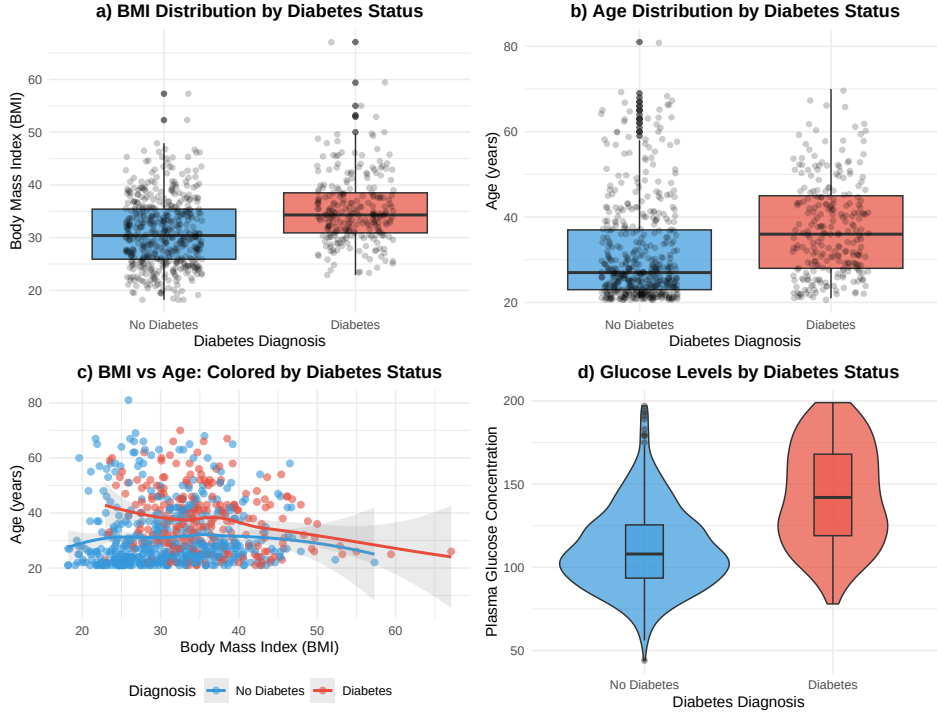


Figure 1: a, b) Distributions of BMI (a) and age (b) by diabetes status, showing higher values in the diabetes group. c) Scatter plot of BMI versus age, colored by diabetes status. Diabetes cases cluster in regions of higher BMI and older age. d) Distribution of plasma glucose concentration by diabetes status, showing clear separation between groups

3 Proposed Method

3.1 Bayesian Logistic Regression Model

We model the binary diabetes outcome $y_i \in \{0, 1\}$ for individual $i = 1, \dots, n$ using logistic regression. Let $\mathbf{x}_i = (x_{i1}, x_{i2}, x_{i3}, x_{i4})^T$ denote the vector of standardized predictors (glucose, BMI, age, blood pressure) for individual i . The model is specified as:

$$y_i \mid \pi_i \sim \text{Bernoulli}(\pi_i) \quad (1)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} \quad (2)$$

where $\pi_i = P(y_i = 1 \mid \mathbf{x}_i, \boldsymbol{\beta})$ is the probability that individual i has diabetes, β_0 is the intercept, and $\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3, \beta_4)^T$ are the regression coefficients for the standardized predictors.

The likelihood function for the full dataset is:

$$L(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \quad (3)$$

where $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)^T$ is the parameter vector.

Because all continuous predictors are standardized to have mean 0 and standard deviation 1, each coefficient β_j represents the change in log-odds of diabetes associated with a one standard deviation increase in predictor j , holding all other predictors constant. The corresponding odds ratio is $\exp(\beta_j)$.

3.2 Prior Distributions

We adopt weakly informative priors for all regression coefficients:

$$\beta_0 \sim \mathcal{N}(0, 5^2) \quad (4)$$

$$\beta_j \sim \mathcal{N}(0, 2.5^2), \quad j = 1, 2, 3, 4 \quad (5)$$

The prior $\mathcal{N}(0, 2.5^2)$ on the standardized coefficients is motivated by Gelman et al. [2] for weakly informative logistic regression priors. For standardized predictors, this prior assigns 95% probability to coefficients between approximately -5 and 5, corresponding to odds ratios between 0.007 and 148. This range covers nearly all clinically plausible effects while preventing unrealistic extreme values. The wider prior $\mathcal{N}(0, 5^2)$ on the intercept reflects greater prior uncertainty about the baseline log-odds of diabetes while still providing some regularization.

3.3 Posterior Distribution

By Bayes' theorem, the posterior distribution is:

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) \propto L(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}) \cdot p(\boldsymbol{\theta}) \quad (6)$$

The posterior does not have a closed-form expression, requiring computational methods for inference.

3.4 Computational Inference: Metropolis MCMC

The Metropolis algorithm generates a Markov chain $\{\boldsymbol{\theta}^{(t)}\}_{t=1}^T$ whose stationary distribution is the posterior. At iteration t :

1. **Proposal:** $\boldsymbol{\theta}^* \sim q(\cdot \mid \boldsymbol{\theta}^{(t)})$
2. **Acceptance probability:** $\alpha = \min\left\{1, \frac{p(\boldsymbol{\theta}^* \mid \mathbf{y}, \mathbf{X})}{p(\boldsymbol{\theta}^{(t)} \mid \mathbf{y}, \mathbf{X})}\right\}$
3. **Accept/reject:** With probability α , set $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^*$; otherwise $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)}$

We use a multivariate normal random-walk proposal:

$$\boldsymbol{\theta}^* = \boldsymbol{\theta}^{(t)} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_5) \quad (7)$$

where $\sigma = 0.12$ is the proposal standard deviation.

After preliminary runs, we selected $\sigma = 0.12$, which yielded an acceptance rate of approximately 22.8%, falling within the theoretically optimal range of 15-40% for random-walk Metropolis in moderate dimensions. We ran the algorithm for $T = 30,000$ iterations, discarding the first $B = 5,000$ as burn-in, retaining 25,000 samples for posterior inference. The chain was initialized at $\theta^{(0)} = \mathbf{0}$. From the retained MCMC samples, we compute posterior means, standard deviations, 95% credible intervals, and odds ratios. We examine convergence diagnostics including traceplots, autocorrelation functions, and effective sample sizes.

4 Data Analysis and Results Interpretation

4.1 Posterior Distribution of Regression Coefficients

We implemented a random-walk Metropolis algorithm with 30,000 total iterations, discarding the first 5,000 iterations as burn-in. The remaining 25,000 post-burn-in samples constitute our posterior inference. Table 2 presents posterior summaries for all regression coefficients.

Table 2: Posterior means, standard deviations, and 95% credible intervals for regression coefficients (log-odds scale)

Parameter	Posterior Mean	Posterior SD	2.5%	97.5%
Intercept	-0.878	0.096	-1.068	-0.688
Glucose (std.)	1.081	0.110	0.868	1.302
BMI (std.)	0.647	0.107	0.439	0.859
Age (std.)	0.409	0.099	0.220	0.606
Blood Pressure (std.)	-0.110	0.104	-0.314	0.093

Because all continuous predictors are standardized to have mean zero and standard deviation one, each coefficient represents the change in log-odds of diabetes associated with a one standard deviation increase in that predictor, holding all other variables constant.

The coefficient for glucose shows the strongest positive association, with a posterior mean of 1.081 and a 95% credible interval of (0.868, 1.302) that lies entirely above zero. This provides strong evidence that higher glucose levels substantially increase diabetes risk. The coefficients for BMI and age are also clearly positive, with 95% credible intervals of (0.439, 0.859) and (0.220, 0.606) respectively, both excluding zero. This indicates robust positive associations between these predictors and diabetes status.

In contrast, the coefficient for blood pressure has a posterior mean near zero (-0.110) with a credible interval of (-0.314, 0.093) that crosses zero. This suggests that blood pressure does not exhibit a strong independent association with diabetes after controlling for glucose, BMI, and age. The credible interval indicates substantial posterior uncertainty about whether blood pressure has any effect at all—it could be weakly negative, essentially zero, or weakly positive.

4.2 Posterior Odds Ratios

While regression coefficients quantify effects on the log-odds scale, odds ratios provide more intuitive clinical interpretation. We exponentiated each posterior sample of the coefficients to obtain posterior distributions for odds ratios. Table 3 presents these results.

Table 3: Posterior odds ratios and 95% credible intervals for standardized predictors

Predictor	Posterior Mean OR	2.5%	97.5%
Glucose (std.)	2.948	2.382	3.677
BMI (std.)	1.910	1.551	2.362
Age (std.)	1.506	1.247	1.833
Blood Pressure (std.)	0.896	0.730	1.097

The odds ratio interpretation is straightforward: for a one standard deviation increase in a predictor, the odds of diabetes are multiplied by the corresponding odds ratio, holding all other predictors constant.

Glucose demonstrates the dominant effect with a posterior mean odds ratio of 2.948 (95% CI: 2.382–3.677). This means that a one standard deviation increase in glucose—approximately 28.1 mg/dL in our sample—is associated with nearly tripling the odds of diabetes. The 95% credible interval indicates we can be quite confident the true odds ratio lies between approximately 2.4 and 3.7, representing a strong and clinically significant effect. This finding strongly aligns with medical knowledge that plasma glucose concentration is a primary diagnostic indicator of diabetes.

BMI shows a moderate positive association with a posterior mean odds ratio of 1.910 (95% CI: 1.551–2.362). A one standard deviation increase in BMI (approximately 6.6 kg/m²) is associated with roughly doubling the odds of diabetes. This effect, while substantial, is weaker than that of glucose. The credible interval is relatively narrow and lies entirely above 1, providing strong evidence for a positive BMI effect. This reflects the well-established connection between obesity and metabolic dysfunction leading to type 2 diabetes.

Age exhibits a modest positive association with a posterior mean odds ratio of 1.506 (95% CI: 1.247–1.833). A one standard deviation increase in age (approximately 11.4 years) is associated with about a 50% increase in diabetes odds. While this effect is smaller than those of glucose and BMI, the credible interval lies entirely above 1, indicating a clear positive association. This is consistent with clinical understanding that diabetes risk increases with age due to cumulative exposure to risk factors and age-related metabolic changes.

Blood pressure shows a posterior mean odds ratio of 0.896 (95% CI: 0.730–1.097), very close to 1 (null effect). Critically, the 95% credible interval spans 1, ranging from 0.730 to 1.097. This indicates substantial posterior uncertainty about the direction and magnitude of any blood pressure effect. The data are consistent with blood pressure having a weak negative association, no association, or a weak positive association with diabetes. After controlling for glucose, BMI, and age, blood pressure does not emerge as a strong independent predictor in this dataset.

4.3 MCMC Convergence Diagnostics

4.3.1 Acceptance Rate

Our random-walk Metropolis algorithm with proposal standard deviation $\sigma = 0.12$ achieved an acceptance rate of:

$$\text{Acceptance Rate} = 0.2281 \approx 22.8\%$$

This acceptance rate lies well within the theoretically optimal 15–40% range recommended for random-walk Metropolis samplers in moderate dimensions, implying that the proposal steps are suitably scaled — neither so large that they cause low acceptance and poor exploration, nor so small that they yield high acceptance but inefficient movement through the posterior. In turn, this indicates efficient exploration of the target distribution and suggests that the chosen proposal standard deviation is appropriately calibrated for this problem.

4.3.2 Traceplots

Figure 2 displays traceplots for the intercept and all four standardized predictor coefficients across all 30,000 MCMC iterations.

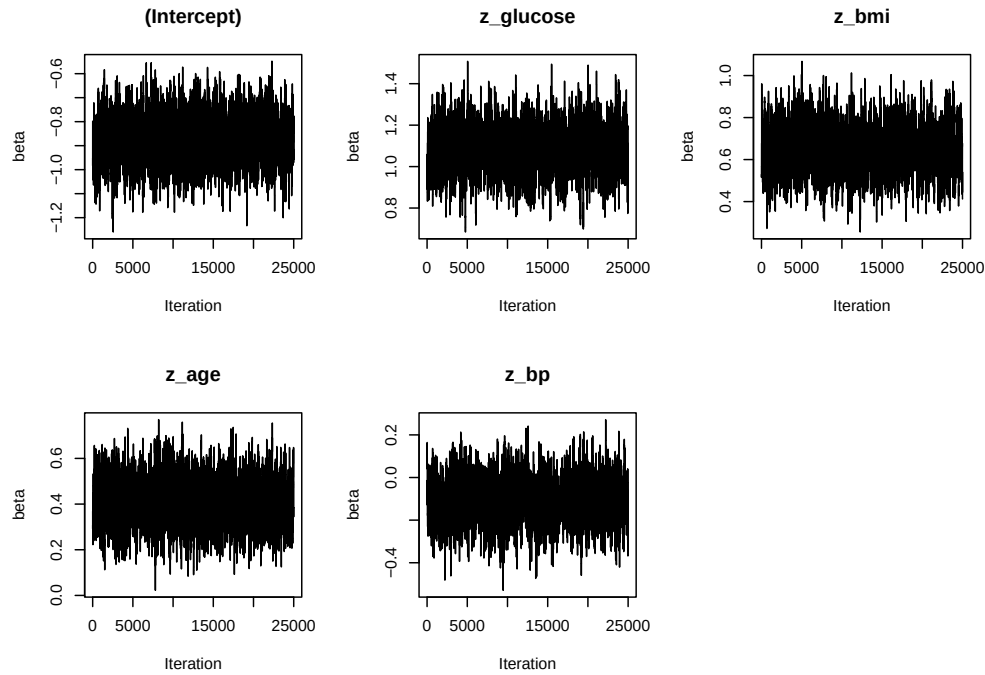


Figure 2: Traceplots showing MCMC samples for all regression coefficients over 30,000 iterations. The first 5,000 iterations (burn-in period) show initial convergence to the stationary distribution, while the remaining 25,000 iterations exhibit stable mixing around constant posterior means.

Healthy traceplots should exhibit several characteristics: rapid oscillation around a stable mean with no visible trends or drift over time, good mixing without the chain getting stuck in local regions, and convergence to the stationary distribution after the burn-in period. All five traceplots in Figure 2 demonstrate these desirable properties.

The glucose and BMI coefficients show particularly tight, rapidly mixing behavior reflecting strong data information for these parameters. The blood pressure coefficient exhibits slightly wider oscillations consistent with its larger posterior uncertainty. Importantly, none of the chains show evidence of slow mixing or failure to explore the posterior space adequately.

4.3.3 Autocorrelation Function

The autocorrelation function (ACF) measures the correlation between MCMC samples separated by different lags. For a well-mixing chain, autocorrelation should decay toward zero as lag increases. While some autocorrelation is inherent in random-walk Metropolis algorithms, excessive persistence would indicate inefficient sampling.

Figure 3 presents ACF plots for all five parameters up to lag 40. The ACF plots exhibit healthy decay: autocorrelation at lag 1 is approximately 0.35–0.45, consistent with expectations for a random-walk Metropolis sampler at this acceptance rate; autocorrelation decreases monotonically with lag for all parameters; by lags 15–20, values are near zero and lie within the 95% confidence bands (blue dashed lines); and no parameters show problematic long-range dependence or periodic patterns.

The intercept and blood pressure coefficients show slightly higher initial autocorrelation, but still decay appropriately. The glucose, BMI, and age coefficients exhibit faster decay, consistent with stronger data information for these parameters. Overall, these ACF plots indicate that our MCMC sampler is mixing adequately and not suffering from excessive autocorrelation that would severely degrade effective sample size.

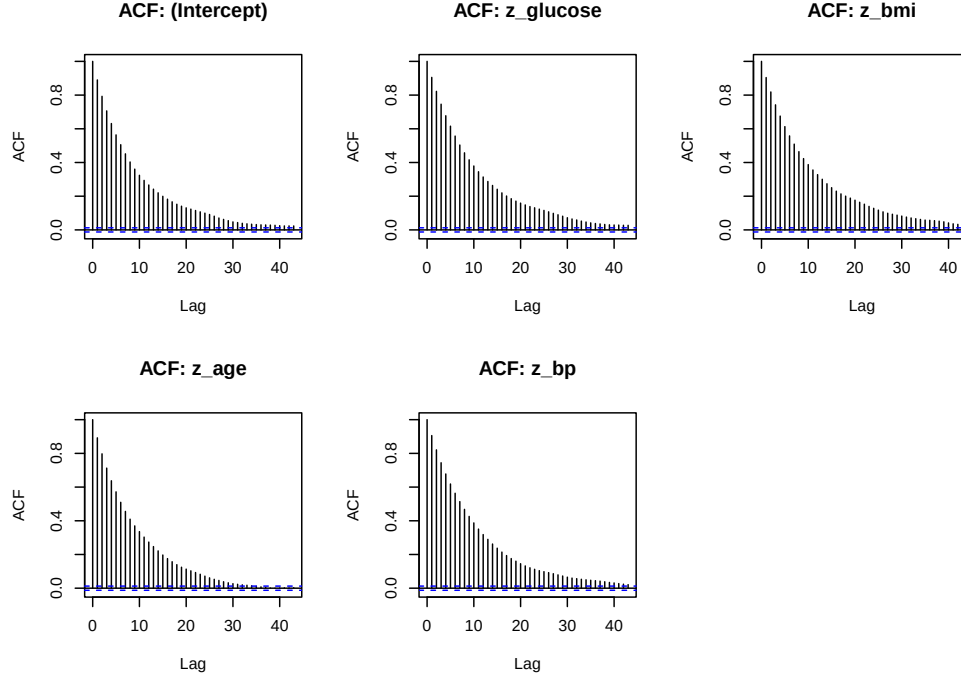


Figure 3: Autocorrelation functions (ACF) for all regression coefficients showing correlation between samples at different lags. All parameters exhibit appropriate decay in autocorrelation, with most correlations becoming negligible by lag 15–20.

4.3.4 Effective Sample Size

While we retained 25,000 post-burn-in MCMC samples, these samples are autocorrelated and thus contain less information than 25,000 independent samples would. The effective sample size (ESS) quantifies the equivalent number of independent samples represented by our correlated MCMC draws.

Table 4 presents ESS values, posterior standard deviations, and Monte Carlo standard errors for all parameters.

Table 4: Effective sample size and Monte Carlo diagnostics for all parameters

Parameter	Posterior SD	ESS	MCSE
Intercept	0.096	1454.9	0.0025
Glucose (std.)	0.110	1204.8	0.0032
BMI (std.)	0.107	1196.2	0.0031
Age (std.)	0.099	1424.2	0.0026
Blood Pressure (std.)	0.104	1185.4	0.0030

All parameters exhibit ESS values exceeding 1,185, representing approximately 5-6% efficiency relative to the 25,000 total post-burn-in samples. These ESS values are quite respectable for a basic random-walk Metropolis sampler and substantially exceed the commonly cited threshold of ESS > 400 for reliable posterior inference.

The Monte Carlo standard error (MCSE) quantifies the uncertainty in our posterior mean estimates due to finite MCMC sampling. All parameters have MCSE values less than 0.0032, which is small relative to the posterior standard deviations (ranging from 0.096 to 0.110). The ratio of MCSE to posterior SD ranges from 2.5% to 3.0%, indicating that Monte Carlo error contributes negligibly to our overall posterior uncertainty. This confirms that our MCMC sample size is adequate for precise posterior estimation.

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